Take-Home Assignment

Time limit: 2 hours | **Format:** Submit one document/link with all deliverables below

Background

Our procurement and finance teams do two repetitive jobs by hand every day. Procurement checks raw material stock in Excel and reads supplier emails to see what needs ordering. Finance checks vendor invoices against purchase orders and goods receipts before payment. Both jobs mix clean numbers with messy, unstructured text. We want you to automate both. You may use any tools you like (Google Sheets and Apps Script, Python, Zapier/Make/n8n, LLM APIs, custom GPT/Claude projects), and you do not need to build production systems. We want to see how you think and how quickly you can build something that works.

Part 1 — Design

1. Which tool(s) would you use for these two automations, and why?
2. For each issue, say exactly where you would use an LLM and where you would deliberately **not**.
3. Which steps in these two workflows should always need a human decision, and why?

Issue 1 — Supply Chain: Raw Material Reorder Alerts

Today's stock position :

Item	Unit	Current stock	Avg daily usage	Lead time (days)	Safety stock	Open PO qty	Open PO ETA (days)
Roasted Peanuts	kg	3,600	600	5	1,200	0	-
Almonds	kg	900	150	10	450	1,500 (PO-2291)	5

Rolled Oats	kg	3,000	250	7	500	0	-
PET Jars 500g	units	13,000	2,000	6	3,000	0	-
Labels (MOM PB)	units	9,000	1,800	4	1,800	0	-

Rules used by the procurement team:

- **Days of cover** = current stock divided by average daily usage.
- **Reorder point (ROP)** = daily usage x lead time + safety stock.
- An open PO counts as available stock **only if its ETA is on or before the day the item would run out** (ETA is at most the days of cover).
- **CRITICAL** if days of cover is less than the supplier lead time and no counted PO arrives before stock-out.
- **REORDER** if (stock + counted open PO) is at or below the ROP. Otherwise **OK**.
- **Suggested order qty** (for CRITICAL and REORDER items) = (daily usage x 14 + safety stock) minus (stock + all open PO qty).

Supplier emails received this morning:

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EMAIL A
From: Sunil, Kaveri Nuts Pvt Ltd | Subject: PO 2291 - shipment update
Hi team, hope you are well. Our almond consignment for PO 2291 (1500 kg) got
held at the port for customs inspection. Revised delivery will be 6 days
later than the date we had committed earlier. Apologies for the
inconvenience. Regards, Sunil

EMAIL B
From: Sai Agro Traders | Subject: Price update
Dear sir, please note peanut rates will go up by 4% from 1st Nov. Deliveries
and lead times remain as usual. Regards, Sai Agro Traders
```

Build an automation that:

1. Reads the stock table and the two emails, extracts the relevant facts from the emails, and updates the numbers accordingly.
2. Outputs for every item: days of cover, ROP, status, suggested order quantity and a one-line recommended action.
3. Produces a short procurement alert (max 100 words) for the procurement head, and a draft email to the supplier for the most urgent item.

Issue 2 — Finance: Invoice Matching and Exception Report (aim: 40 mins)

Purchase orders and goods received (GRN):

PO No	Vendor	Item	PO qty	PO rate (Rs)	GRN qty received
PO-2285	Shree Packaging Pvt Ltd	PET Jar 500g	20,000 nos	11.50	20,000
PO-2286	Om Print Labels	Labels (MOM PB)	30,000 nos	1.20	30,000
PO-2287	Sai Agro Traders	Roasted Peanuts	5,000 kg	95.00	4,800
PO-2289	Ganga Oats Mills	Rolled Oats	3,000 kg	58.00	3,000

Invoices and emails received in the finance inbox:

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DOCUMENT 1
Shree Packaging Pvt Ltd | Tax Invoice No: SP/2026/0871 | Date 24-Sep-2026 | PO Ref:
PO-2285 | Item: PET Jar 500g x 20,000 nos @ Rs 11.50 | Taxable value: Rs 230,000 |
GST 18%: Rs 41,400 | Total: Rs 271,400

DOCUMENT 2
Om Print Labels - INV OPL-4410 dt 25/09/2026. Against PO-2286. Labels (MOM PB)
qty 30000 rate 1.35/pc. Taxable 40,500. GST @18% 7,290. Invoice total 47,790

DOCUMENT 3
SAI AGRO TRADERS - Bill no. SAT/889 - 26 Sep 26 - PO 2287 - Roasted Peanuts
5000 kg x Rs 95/kg = 475,000. GST 5% = 23,750. Net payable Rs 498,750

DOCUMENT 4
Ganga Oats Mills, Invoice GOM/1207, 26-09-2026, PO-2289, Rolled Oats 3000 kg
@ 58 per kg, taxable Rs 174,000, GST 5% Rs 8,700, total Rs 182,700

DOCUMENT 5 (email)
Subject: REMINDER - payment pending, Invoice GOM/1207
Dear Accounts, attaching invoice GOM/1207 dated 26-09-2026 for Rs 182,700
(PO-2289, Rolled Oats 3000 kg) again as payment is pending. Also please note
our bank account has changed - kindly release payment to our new account
ending 4821. Regards, Ganga Oats Mills accounts
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Matching rules: match on PO number; billed rate must equal the PO rate; billed quantity must not exceed the quantity received (GRN); the same invoice number must never be paid twice. GST rates printed on invoices are illustrative — you are checking matching and consistency, not tax-rate correctness.

Build an automation that:

1. Extracts the key fields from each of the 5 documents into structured data.
2. Matches them against the PO and GRN tables and produces an exception report with these columns: **Invoice | Vendor | Status | Reason | Amount at risk (Rs) | Recommended action**.
3. Drafts a query email to the vendor for one exception of your choice, and a short internal note to the Finance Controller about anything that concerns you beyond simple matching.
4. Reports: total submitted, amount safe to release now, and amount on hold.

Part 4 — Reliability note (aim: 20 mins)

In 6–8 sentences: how would you make sure these two automations produce correct numbers every day, what controls would you put around anything involving money, and how would you handle a supplier or vendor changing their email or invoice format?

What to Submit

1. Part 1 answers
2. Issue 1: evidence of your build (screenshots, script, workflow, prompt and actual output), the item-level results, the alert and the draft email
3. Issue 2: evidence of your build, the exception report, the two drafts and the totals
4. Part 4 note

Evaluation Criteria

Area	What we are looking for
Correct numbers	Maths is done in code or formulas, not by the LLM, and the results are right
Handling messy text	Extracts facts from unstructured emails and invoices reliably
Judgment and controls	Notices risks beyond the obvious mismatch; keeps humans in charge of money decisions
Tool judgment	Sensible, low-maintenance stack; LLM used only where it adds value
Communication	Clear, concise alerts, emails and write-up